25900117 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM100 - INTELLIGENT AUTOMATION TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which type of machine learning involves an agent learning through rewards and penalties? [K] / [CO1]

   a). Supervised Learning b). Unsupervised Learning

   c). Semi-supervised Learning d). Reinforcement Learning

2. Which pandas function is used to read data from an SQL database? [U] / [CO2]

   a). pd.fetch_sql() b). pd.import_sql() c). pd.load_sql() d). pd.read_sql()

3. Which of the following is a classification algorithm? [K] / [CO3]

   a). K-Means Clustering b). Apriori Algorithm

   c). PCA d). Logistic Regression

4. What are the main components of a neural network? [U] / [CO1]

   a). Input layer, output layer, and b). Input layer and output layer c). Nodes and weights only d). Weights and biases only

hidden layers only

5. Which of the following is an advantage of KNN? [K] / [CO3]

a). Fast for large datasets b). Does not require labeled data

c). Requires a complex training phase d). Simple and easy to implement

6. Which of the following is a common challenge in machine learning? [U] / [CO1]

a). Lack of data for training b). Overfitting the model to the training data c). Computational power limitations d). All of the above

7. What is the purpose of momentum in gradient descent? [K] / [CO2]

a). To slow down convergence b). To avoid local minima c). To accelerate convergence and reduce oscillations d). To make the gradient larger

8. Which of the following best describes reinforcement learning? [U] / [CO3]

a). Learning from labeled datasets b). Learning without feedback

c). Learning through rewards and penalties d). Learning based on statistical analysis

9. Which type of Machine Learning involves labeled data? [K] / [CO3]

a). Supervised Learning b). Unsupervised Learning

c). Reinforcement Learning d). Deep Learning

10. Which of the following is an advantage of artificial intelligence? [An] / [CO1]

a). Reduces the time taken to solve the problem b). Helps in providing security c). Have the ability to think hence makes the work easier d). All of the above

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. What are some real-world applications of AI? [K] / [CO1]

12. What is the default file extension for Python scripts? [U] / [CO2]

13. What are some real-world applications of Unsupervised Learning techniques? [A] / [CO3]

14. What are variables in Python? Give an example. [U] / [CO2]

15. What is the difference between agglomerative and divisive Hierarchical Clustering? [K] / [CO3]

16. What are some applications of Machine Learning in the financial industry? [A] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the steps involved in connecting to a database and extracting data using Python. Provide sample code to illustrate your explanation. [K] / [CO2]

18. Explain the concept of regression analysis with a simple example. [K] / [CO3]

19. Compare and contrast supervised learning, unsupervised learning and reinforcement learning. [An] / [CO1]

20. How do you debug Python code in Jupyter Notebook? Explain with examples. [U] / [CO2]

[10x2 = 20]